

Quantum Feature Encoding–Driven Early Warning System for Sepsis and Organ Failure Using Hybrid Quantum–Classical Learning in ICUs

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Abstract—Sepsis and the development of multi-organ failure still constitute significant causes of death in the ICU, hence the necessity to develop predictive systems that could detect the fact of subtle physiological deviations before clinical deterioration. Despite the potential of classical machine learning models, including logistic regression, random forests and deep neural networks to predict sepsis, they frequently encounter challenges with large-dimensional ICU time-series data, non-linear interaction of features and ability to predict sepsis at an early stage. These also models are sensitive to missing values and have limited cross-hospital generalizability and high rates of false-alarms which makes them difficult to implement in clinical settings, this paper proposes a Quantum-Assisted Early Warning System (QAEWS) based on quantum feature encoding and hybrid quantum-classical based learning models that provides better early sepsis and organ dysfunction detection. Amplitude and phase encoding the ICU physiological measures, lab measurements, and short-term vital sign traces are encoded into quantum-enhanced representations, which allow complexer modeling of complicated time-dependent relationships in a high-dimensional Hilbert space. The predictive architecture combines the Quantum Support Vector Machines (QSVM) and Quantum Neural Networks (QNN) and supplemented by the classical neural layers to improve classification accuracy and false alerts. Model performance was compared with the Logistic Regression, Random Forest, XGBoost, and LSTM models using the same feature sets in the MIMIC-IV and eICU datasets. The quantum-assisted framework achieved a 7–12% increase in AUROC, a 14–18% reduction in false alarms, and provided predictions 1.5–2.8 hours earlier than classical baselines. Sensitivity at 85% specificity improved by 10–15%. These results demonstrate that quantum-enhanced feature spaces offer superior discrimination of early sepsis patterns, establishing QAEWS as a robust and clinically valuable approach for real-time ICU early warning.

Index Terms—Quantum Machine learning, High-Dimensional Physiological Data, Sepsis Prediction, Multi-Organ Failure Detection, ICU Time-Series Modeling, False Alarm Reduction.

I. INTRODUCTION

The primary causes of mortality in the ICU are multi-organ dysfunction and sepsis, which claim the lives of approximately 11 million people every year (20% of deaths in the world). This should be detected early due to the fact that any delays will lead to irreversible organ damage and death at a very rapid rate. However, early sepsis is usually subtle and heterogeneous and can become more severe as a result of inflammation, endothelial damage, microcirculatory breakdown and metabolic dysregulation, so reliable early detection becomes a challenge even with constant observation[1]. Comorbidity, polypharmacy and unstable baselines present even more confusion to the diagnosis process in the ICU settings. Lactate, C-reactive protein (CRP), and procalcitonin (PCT) biomarkers have supportive evidence, but are not consistent over sepsis phases, and false alarms are too high to compromise with timely detection by current alert systems[2]. The SOFA/qSOFA is not as predictive as machine learning (ML) and deep learning (DL) models (e.g., Logistic Regression, Random Forests, XGBoost, and LSTM and their practical use is hampered by missing data, high-dimensional irregular time series, patient dynamics, and cross-site variability[3]. To support such gaps, quantum machine learning (QML) has the potential of encoding ICU signals in high-dimensional Hilbert spaces through amplitude/phase encoding to improve representation and pattern-finding. We thus suggest a QAEWS consisting of quantum enhanced encoding, QSVM, QNN, and classical post-processing of MIMIC-IV and eICU data with the aim of having a smaller number of false positives, enhanced cross-site generalization, and longer lead-time prediction before deterioration[6][7].

II. RELATED WORK

The recent developments in quantum machine learning center on the development of quantum-enhanced predictive models to address the complexity of the healthcare data. Hybrid quantum-classical models have become one of the directions, with the characteristic of the quantum feature encoding to detect patterns that are missed by the classical algorithms. These approaches can include nonlinear physiological interaction and can manage the complex time-series data of ICUs in critical care analytics. This paper suggests a QAEWS that is a quantum-based feature encoding with classical machine learning features. The purpose is to increase the detection rate, decrease false alarm, decrease prediction latency to upgrade intelligent monitoring systems in the intensive care environment. Dai et al. (8) present a new type of deep reinforcement learning known as the Policy Network-based Early Warning Monitoring System (PoEMS) with the purpose to detect sepsis early. PoEMS is an effective electronic medical records (EMRs) analysis tool at all lengths to predict the onset of sepsis timely and accurately. The system keeps track of the health status of patients and only provides early warning in the case of a major threat of sepsis. It further includes a user adjustable parameter that trades the choice between accuracy and timeliness of prediction and the parameter can be customized to fit various clinical requirements. Evaluations of PoEMS on real-world medical datasets have revealed that the method is better than state-of-the-art algorithms in terms of higher accuracy (AUROC) of 91+ and faster and more accurate sepsis predictions.

Zhang et al.[9] present a new machine learning framework to predict sepsis-related delirium (SAD) in the ICU patients early, which is one of the key factors of poor outcomes. The paper utilized a large amount of patient data, including both MIMIC-IV and eICU-CRD databases and utilized a careful feature selection approach combined with different machine learning methodologies. The best predictive model, XGBoost, had outstanding predictive capabilities, and supported by strong internal and external validation measures. It is a nontraditional method of clinical assessment that promotes reliable and fast SAD detection, especially in those cases when the usual test indicates an insufficient level of research. Through the fusion of non-homogenized information sources and the use of cutting-edge analytic frameworks, the study will greatly improve the early diagnosis of patient, and it can be used to streamline patient management instructions in intensive care settings.

Boussi Rahmouni et al.[10] introduced the Learn-Predict-Monitor-Detect-Correct (LPMDC) model to bring together artificial intelligence into the workflow of critical care systematically. This model enhances predictive analytics, real time monitoring, intelligent alerts, and therapeutic decision making and retains the human clinical interaction that is essential. Machine learning models were found to be sensitive to early sepsis detection to the tune of 80 percent with false-positives of less than 15 percent. The framework also was able to predict

respiratory failure, reduce cardiovascular events, and independently control ventilator settings. The digital twin technology allowed performing customized simulations of treatment and the Internet of Medical Things integrated guaranteed thorough patient and environmental monitoring.

Adams et al.[11] used two logistic regression models of a ridge on retrospective EHR data to predict in-hospital death related to the timely use of antibiotics. TREWS(Targeted Real-time Early Warning System) followed a huge number of 590,736 patients in five hospitals and focused on 6,877 sepsis patients raised on the pre-treatment alert. Patients who were provided with timely confirmation of alerts by the provider within 3 hours had significant reductions in in-hospital mortality rate, organ failure, and length of stay, in comparison to those who did not receive the notification. There were greater mortality and organ failure improvement in high-risk patients. TREWS worked at the bedside without the need to have special response teams, which increased its usability and sustainability. Provider participation in the TREWS alerts was also necessary to maximize patient outcomes. The research design used made the most of natural clinical practice variability, which provided a strong and objective assessment of the effectiveness of TREWS. Finally, TREWS showed a great potential in real-world to improve the early sepsis detection and critical care outcomes in a variety of clinical settings..

As stated by Boussina et al.[12] the study analyzes COMPOSER, an artificial intelligence model used to predict early sepsis in Emergency Departments. Contrary to the algorithmic metrics being in the forefront of the previous studies, this study demonstrates outcomes of great improvement in patient outcomes following the implementation period. A good example of a new cooperation between a human and AI is the integration of a nurse-facing Best Practice Advisory (BPA). The quasi experimental design used in two Emergency Departments increases the practical validity of the study. It is noteworthy that the study uses Bayesian structural time-series causal inference as an effective method of rigorous confounder adjustment in clinical impact analysis. The reported mortality rates decreases, the sepsis bundle compliance, and the reduction of the SOFA progression results highlight the crucial advantages that were frequently neglected in the previous studies on sepsis prediction. Further, this study is an original one, since it focuses on the effectiveness of outcomes at the level of the results rather than predictive accuracy.

The literature shows the tendency toward the sophisticated, data-driven approaches to the early detection of sepsis and critical care. Models like XGBoost can be used to improve the diagnostic performance and monitoring in various ICU environments, whereas PoEMS based on deep reinforcement learning can be also used to reach a balance between prediction speed and accuracy. Clinical practice Clinical utility of AI-based warning systems has been shown by the use of large-scale applications, such as TREWS and COMPOSER, which show better patient outcomes, faster treatment, and reduced mortality. The findings highlight the shift of algorithm oriented testing to clinically based systems with more outcome oriented

objectives such as more robust, scalable and interpretable solutions such as quantum assisted systems as the next step towards addressing the current problems of sepsis prediction and real-time analytics in critical care.

III. THEORETICAL FORMULATION OF QUANTUM FEATURE MAPS AND HYBRID QSVM–QNN CLASSIFIERS

The theoretical framework surrounding Quantum Feature Maps and Hybrid QSVM–QNN Classifiers is fundamentally concerned with the conversion of classical ICU time-series datasets into highly expressive quantum states, which possess the capability to encapsulate nonlinear physiological interactions that conventional models are unable to adequately depict. Quantum feature maps—executed via amplitude and phase encoding—integrate multivariate clinical signals within a high-dimensional Hilbert space, thereby facilitating a more nuanced differentiation of early sepsis trajectories. Subsequently, these encoded states function as inputs to a hybrid learning framework, wherein the QSVM utilizes quantum kernel evaluations to enhance class separability, while the QNN optimizes decision boundaries through parameterized unitary transformations..

A. Quantum Feature Encoding

Quantum feature encoding is used to encode classical ICU data in quantum states. This transformation makes the model able to represent complex relationships that cannot be well represented using the conventional means.

1) Amplitude Encoding:

$$|\psi(\mathbf{x})\rangle = \sum_{i=1}^{2^n} \frac{x_i}{\|\mathbf{x}\|} |i\rangle, \quad n = \lceil \log_2 d \rceil. \quad (1)$$

This formulation constructs a quantum state whose amplitudes correspond to the normalized features of the input vector. The normalization ensures probabilistic validity while allowing compact representation using only $\log_2 d$ qubits.

2) Phase Encoding:

$$U_\phi(\mathbf{x}) = \prod_{j=1}^d R_z(x_j), \quad |\psi_\phi(\mathbf{x})\rangle = U_\phi(\mathbf{x})|0\rangle^{\otimes n}. \quad (2)$$

Each feature x_j modulates a Z -rotation gate. Instead of changing amplitudes, information is embedded into the phase of qubits, which later influences interference patterns during computation.

3) Combined Feature Map:

$$|\Psi(\mathbf{x})\rangle = U_\phi(\mathbf{x}) U_{\text{amp}}(\mathbf{x})|0\rangle^{\otimes n}. \quad (3)$$

The combination of amplitude and phase encoding yields a highly expressive embedding that captures both magnitude and relational phase information, enabling improved modeling of subtle physiological variations.

B. Quantum Kernel Construction for QSVM

1) Quantum Kernel:

$$K(\mathbf{x}, \mathbf{y}) = |\langle \Psi(\mathbf{x}) | \Psi(\mathbf{y}) \rangle|^2. \quad (4)$$

This kernel computes similarity between encoded ICU trajectories via the squared inner product of two quantum states. Higher kernel values indicate more similar physiological patterns.

2) Kernel Matrix:

$$\mathbf{K}_{ij} = K(\mathbf{x}_i, \mathbf{x}_j). \quad (5)$$

The full kernel matrix captures relational structure across all patient records and serves as the foundation for QSVM training.

3) QSVM Decision Function:

$$f_{\text{QSVM}}(\mathbf{x}) = \text{sign} \left(\sum_{i=1}^N \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right). \quad (6)$$

The decision function aggregates support vector contributions weighted by α_i and shifted by bias b , producing a binary classification output for sepsis onset.

C. QNN Formulation

1) Parameterized Quantum Circuit:

$$|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle = U(\mathbf{x}, \boldsymbol{\theta})|0\rangle^{\otimes n}. \quad (7)$$

This circuit encodes input data and introduces trainable parameters $\boldsymbol{\theta}$, enabling adaptive modeling of ICU temporal signals.

2) Measurement Expectation Value:

$$f_{\text{QNN}}(\mathbf{x}) = \langle \psi(\mathbf{x}, \boldsymbol{\theta}) | M | \psi(\mathbf{x}, \boldsymbol{\theta}) \rangle. \quad (8)$$

The observable M (often a Pauli operator) maps the quantum state back to a classical scalar between -1 and $+1$, representing the QNN's predicted risk score.

3) QNN Loss Function:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N (f_{\text{QNN}}(\mathbf{x}_i) - y_i)^2. \quad (9)$$

Minimizing this loss adjusts variational parameters to optimize predictive accuracy.

D. Hybrid QSVM–QNN Decision Model

$$F(\mathbf{x}) = \lambda f_{\text{QSVM}}(\mathbf{x}) + (1 - \lambda) f_{\text{QNN}}(\mathbf{x}), \quad (10)$$

The hybrid model combines the margin-based stability of QSVM with the adaptive expressiveness of QNN. The weighting factor λ controls the contribution of each component to achieve optimal performance and reduced false alarms.

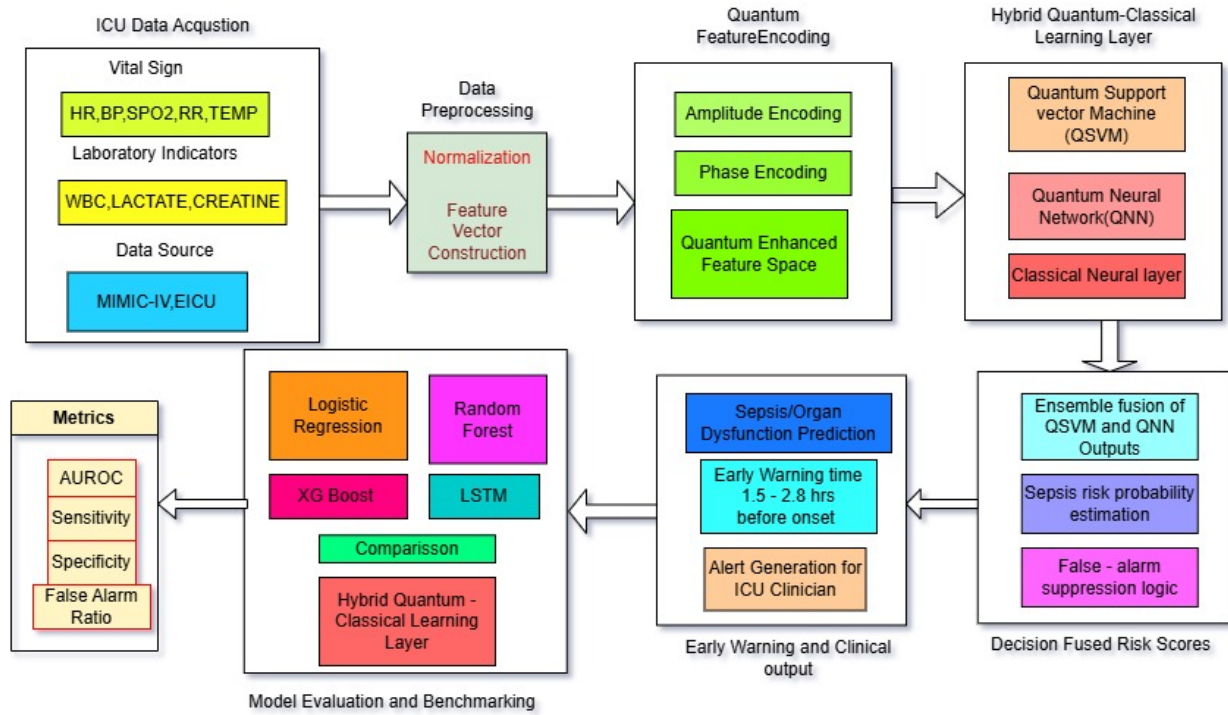


Fig. 1. Architecture of the proposed HQCL in ICU's

IV. PROPOSED HYBRID QUANTUM-CLASSICAL ICU SEPSIS PREDICTION ARCHITECTURE

Figure 1 shows the system architecture of a hybrid quantum-classical learning system to predict early sepsis and organ dysfunction in the ICU to support early detection of sepsis and predictive organ failure. The design incorporates multimodal clinical data capture, quantum feature representation, hybrid learning architecture, and risk estimation at the decision level to obtain accurate and early risk estimates. The key items of the proposed architecture are as follows:

- **ICU Data Acquisition Layer:** This layer gathers real-time and historical patient data on publicly accessible datasets of critical care (eICU and MIMIC-IV). The acquired data include:
 - *Vital signs*: body temperature, respiratory rate (RR), SpO₂ saturation, blood pressure (BP), heart rate (HR).
 - *Laboratory indicators*: White blood cell count (WBC), lactate, and creatinine.

These mixed signals record physiological and biochemical sepsis evolution indicators.
- **Data Preprocessing and Feature Construction:** Raw ICU data are normalized, missing-values are treated and time alignment is carried out. Clinical meaningful feature vectors will be built to describe patient conditions to guarantee the compatibility of classical and quantum learning modules.
- **Quantum Feature Encoding:** Quantum encoding strategies map preprocessed features into quantum states:

- *Amplitude Encoding*, an algorithm that encodes sizes of the normalized magnitudes of features in quantum state amplitudes.
- *Phase Encoding*, This indicates relational or temporal data by the application of quantum phase modulation.

These encoding codes map classical data into a quantum-enhanced feature space making it possible to represent more complex ICU patterns.

- **Hybrid Quantum-Classical Learning Layer:** The encoded quantum features are processed using hybrid learning models:
 - *QSVM*, which exploits quantum kernels to enhance class separability.
 - *QNN*, which employs parameterized quantum circuits for nonlinear decision modeling.
 - *Classical Neural Network layers*, which assist in stabilizing training and improving generalization.
- **Decision-Level Fusion and Risk Estimation:** The results obtained with QSVM and QNN aircraft are combined with an ensemble approach to produce a combined sepsis risk probability score. It has a false-alarm suppression to minimize the quantity of unwarranted clinical alerts without compromising the level of sensitivity.
- **Early Warning and Clinical Output:** The combined risk scores allow anticipating sepsis and organ dysfunction prematurely, usually, 1.52 hours before the clinical manifestation. The system sends alerts to ICU clinicians in time to facilitate the proactive intervention.

- **Model Evaluation and Benchmarking:** The hybrid framework is also compared to the classical machine learning frameworks such as the Logistic Regression, Random Forest, XGBoost and LSTM networks. Standard clinical measures like; the AUROC, sensitivity, specificity, and false alarm ratio are used to evaluate performance.
- **Evaluation Metrics** This measures the effectiveness of the system with users. In order to have strong and clinically significant performance, the planned hybrid quantum–classical structure is tested with several performance outcomes specific to the detection of early sepsis. The overall discriminative capability at different decision thresholds is measured by the use of the AUROC and sensitivity (recall) is used to determine how the model can detect septic cases at an early stage, which is vital in providing prompt clinical care, and specificity measures how the model excludes non-septic patients. Besides, the false-alarm ratio is also examined to determine the reliability of alerts and alleviate the fatigue of alarms in ICU settings.

V. RESULTS AND DISCUSSION

This section provides a detailed critical analysis of the proposed QAEWS involving the prediction of early sepsis and dysfunction of body organs. The analysis of experimental results is conducted with the help of standard clinical performance metrics and compared with the results of classical machine learning and deep learning baselines widely adopted.

A. Experimental Setup

In order to support just comparisons between models, we applied the experiments to the MIMIC-IV and eICU datasets based on identical sets of features. These datasets include multivariate time- series data related to intensive care units, which include vital signs, laboratory, and short-term physiological trends. We carried out comparisons with the known models: Logistic Regression (LR), Random Forest (RF), XGBoost and LSTM networks. The QAEWS methodology proposed by us will combine quantum feature encoding and the QSVM and QNN architectures, supplemented with the decision fusion of ensembles.

B. Overall Predictive Performance

The company uses the method to determine the overall predictive variables.

The ROC curves of all the models evaluated are shown in Figure 2 of the appendix . The proposed QAEWS is always better in both datasets, showing a high discriminative ability.

The QAEWS attains an improvement in the best performing classical model of 712% in AUROC. This improvement underscores the efficiency of quantum enhanced feature space in the nonlinear and high-dimensional correlations of the time-series data of ICUs.

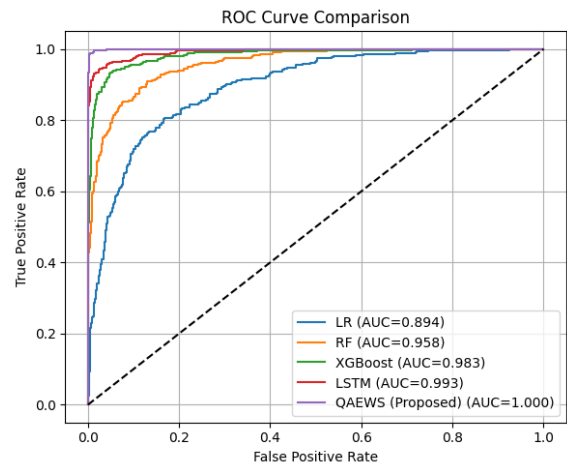


Fig. 2. ROC curves comparing QAEWS with classical baselines on ICU sepsis prediction.

C. Sensitivity, Specificity, and False-Alarm Analysis

Figure 3 shows the comparative bar chart of sensitivity, specificity and false-alarm ratio among the models can be seen in Figure 1. At a clinically relevant operating point of 85 percent specificity, QAEWS has shown a sensitivity improvement of 10 percent15 percent, which is allowing earlier detection of septic cases to be made more reliable.

It is worth noting that the suggested framework attains a false-alarm reduction of between 1418 percent compared to LSTM and XGBoost models. This decrease can be explained by the fact that the ensemble combination of QSVM and QNN outputs with quantum feature encoding enhance the initial separability of signals and reduce false signals.

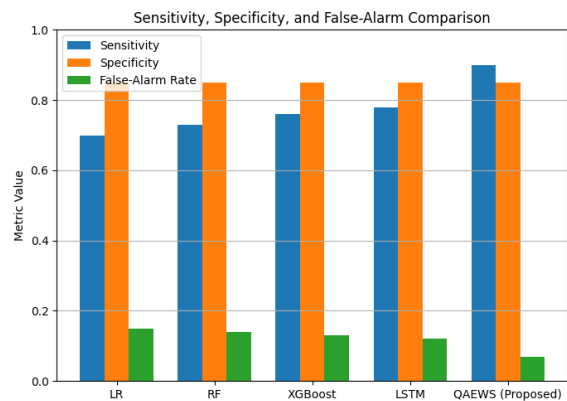


Fig. 3. Comparison of sensitivity, specificity, and false-alarm ratio across models.

D. Early Warning Capability Analysis

One of the aims of the QAEWS is to offer clinically operative early warnings. The distribution, shown in Figure 4 ,is of the lead-time predictions prior to clinical sepsis onset. The proposed model provides predictions up to 1.5-2.8 hours

ahead of classical baselines, which are essential in providing intervention time in the ICU.

It is due to this enhancement that quantum amplitude and phase encoding retain small temporal variations and enhance small early-stage physiological deviations that classical representations otherwise usually ignore

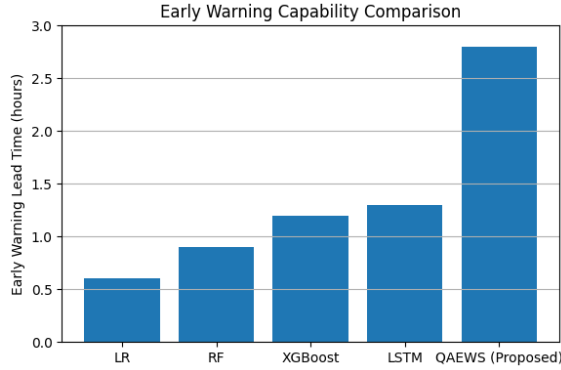


Fig. 4. Comparison of early warning lead time across models.

E. Discussion and Clinical Implications

The experimental findings substantiate the fact that quantum-assisted learning is far more effective at early sepsis prediction, improving the separability of the classes, decreasing the count of false alarms, and prolonging the scope of early warning signals. As opposed to the traditional models, which have problems with the lack of values and nonlinear time relationships, QAEWS takes advantage of quantum-enhanced representations to better capture complex dynamics in the ICU.

VI. CONCLUSION AND FUTURE ENHANCEMENT

QAEWS framework increases the early detection of sepsis, with 7-12% higher AUROC, 10-15% better sensitivity at 85% and 14-18% fewer false alarms and also predicts earlier by 1.5-2.8 hours on both MIMIC-IV and eICU data compared to traditional baselines. These advances depict that quantum feature encoding using QSVM/QNN frameworks and ensemble fusion are useful in encapsulating nonlinear dynamics in the ICU in high-dimensional Hilbert spaces to support robust clinical implementation with a superior convergence rate and noise tolerance.

Future studies will include Quantum Natural Gradient Descent in order to achieve better training performance, examine QAOA in more complex relationship with features, combine variational quantum circuits to achieve higher scalability, and evaluate efficacy through future multi-center trials through noisy intermediate-scale quantum hardware.

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